Internship project report

Title:
Student Performance Data Visualization using Python

Submitted by:
Yashvi chauhan

Submitted to:
Naviotech Solution Pvt. Ltd

Internship Domain:
Data Visualization / Data Analysis

Tools Used:

- Python
- Jupyter Notebook

Year: 2026

2. Certificate Page

CERTIFICATE

This is to certify that **Yashvi** has successfully completed the internship project titled **"Student Performance Data Visualization using Python"** under the guidance of **Naviotech Solution Pvt. Ltd**.

The project work submitted is a genuine record of work carried out during the internship period.

Signature
Academic Head

3. Acknowledgement

I would like to express my sincere gratitude to **Naviotech Solution Pvt. Ltd** for providing me with the opportunity to complete my internship and gain practical experience.

I also thank my mentors and instructors for their valuable guidance and support throughout the project. Their advice helped me understand the concepts of data analysis and visualization using **Python**.

Finally, I would like to thank everyone who supported me during the completion of this project.

4. Table of Contents

1. Introduction
2. Objectives of the Project
3. Tools and Technologies
4. Dataset Description
5. Methodology
6. Data Visualizations
7. Results and Analysis
8. Conclusion
9. Future Scope
10. References

5. Introduction

Data visualization is the process of representing data in graphical form such as charts, graphs, and plots. It helps users understand patterns, trends, and relationships in datasets.

In educational systems, large amounts of data are generated regarding student performance, attendance, and study habits. Analyzing this data using visualization techniques helps identify important insights.

This project focuses on analyzing student performance data using **Python** and visualizing it using different graphical techniques in **Jupyter Notebook**.

6. Objectives of the Project

The objectives of this project are:

- To analyze student performance data
- To create visual representations of academic performance
- To identify patterns between study hours and marks
- To analyze the impact of attendance on performance
- To demonstrate the use of Python libraries for data visualization

7. Tools and Technologies Used

Python

Python is widely used for data analysis and visualization.

Jupyter Notebook

Jupyter Notebook provides an interactive platform to write and execute Python code.

Pandas

Pandas is used for data manipulation and analysis.

Matplotlib

Matplotlib helps create various types of graphs.

Seaborn

Seaborn is used for advanced statistical data visualization.

8. Dataset Description

The dataset used in this project contains information about student academic performance.

Attributes included in the dataset:

- Student Name
- Maths Marks
- Science Marks
- English Marks
- Attendance Percentage
- Study Hours per Week

This dataset helps analyse the relationship between different academic factors.

[1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

data = {
    "Name": ["Aarav","Diya","Rohan","Meera","Kabir","Ananya","Vivaan","Isha","Aditya","Sara"],
    "Maths": [85,78,92,70,88,95,60,80,76,89],
    "Science": [80,74,90,65,84,93,58,79,72,87],
    "English": [75,82,88,72,85,91,62,77,70,90],
    "Attendance": [92,85,96,78,88,98,70,86,80,94],
    "Study_Hours": [15,12,18,10,14,20,8,13,11,17]
}

df = pd.DataFrame(data)
df
```

[1]:

	Name	Maths	Science	English	Attendance	Study_Hours
0	Aarav	85	80	75	92	15
1	Diya	78	74	82	85	12
2	Rohan	92	90	88	96	18
3	Meera	70	65	72	78	10
4	Kabir	88	84	85	88	14
5	Ananya	95	93	91	98	20
6	Vivaan	60	58	62	70	8
7	Isha	80	79	77	86	13
8	Aditya	76	72	70	80	11
9	Sara	89	87	90	94	17

9. Methodology

The project was completed using the following steps:

1. Creating the dataset containing student information
2. Importing the dataset using the Pandas library
3. Performing data analysis
4. Applying visualization techniques using Matplotlib and Seaborn
5. Interpreting the results and identifying patterns

10. Data Visualizations

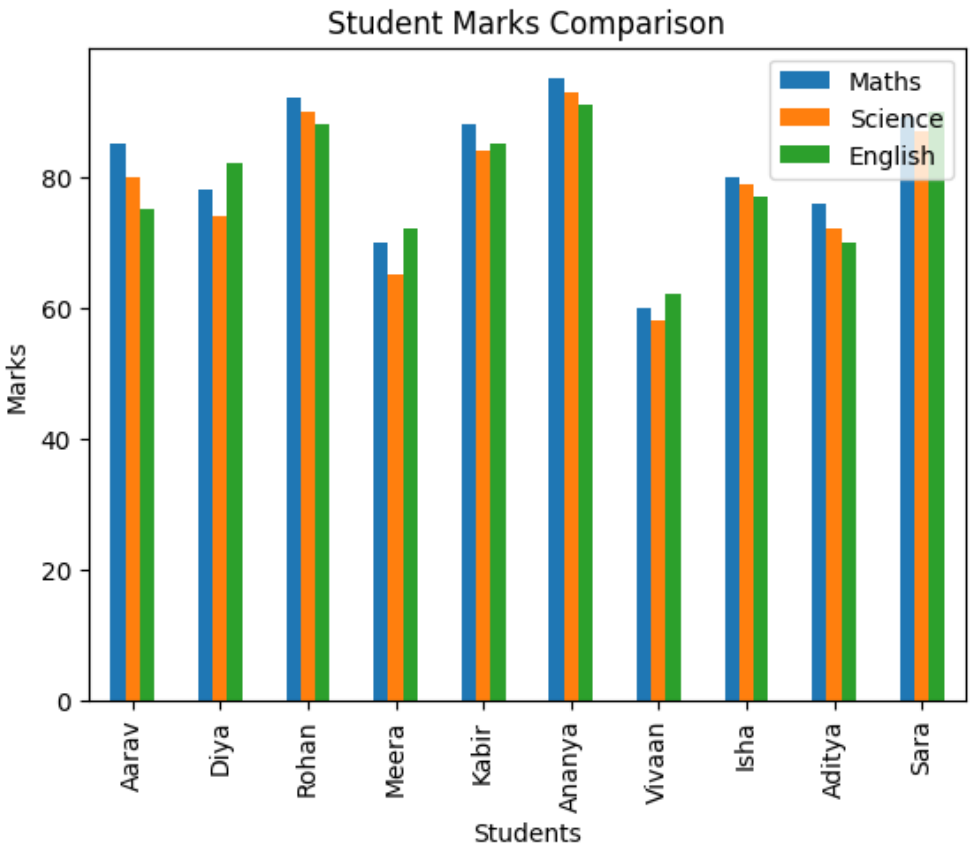
This project includes the following visualizations:

Bar Chart

Comparison of marks across subjects.

```
[2]: df.plot(x="Name", y=["Maths","Science","English"], kind="bar")

     plt.title("Student Marks Comparison")
     plt.xlabel("Students")
     plt.ylabel("Marks")
     plt.show()
```

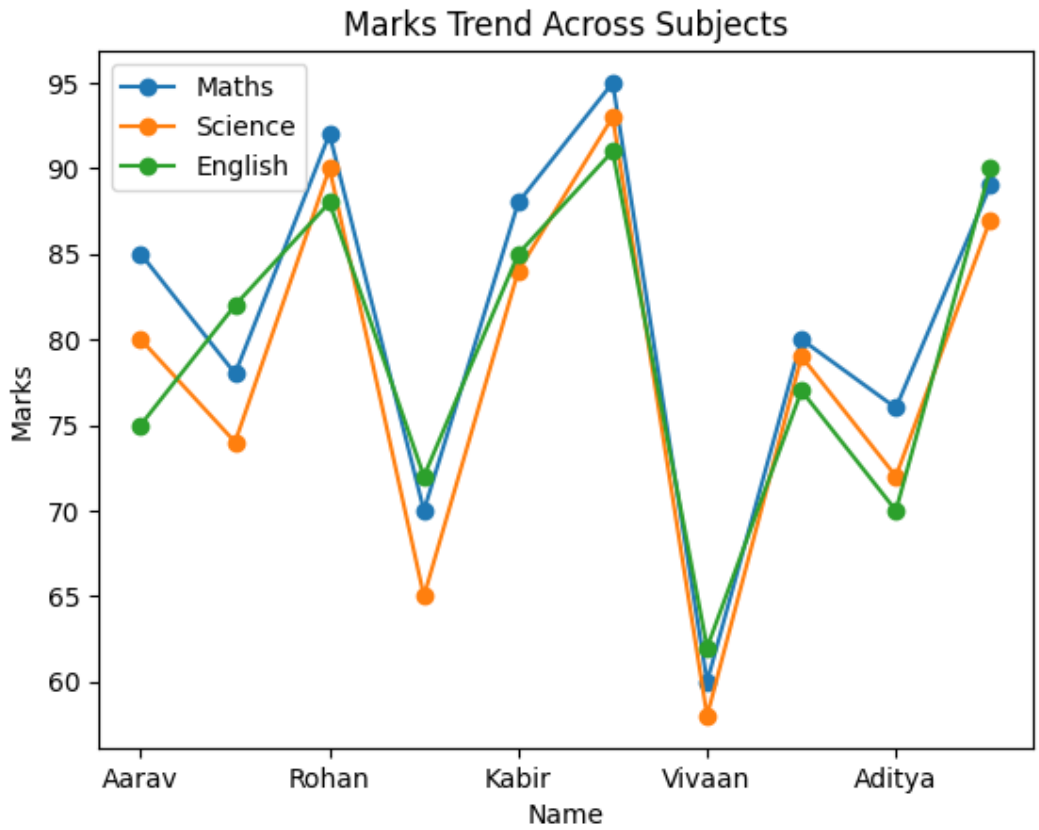


Line Graph

Trend of marks in different subjects.

```
[3]: df.set_index("Name")[["Maths","Science","English"]].plot(marker="o")

     plt.title("Marks Trend Across Subjects")
     plt.ylabel("Marks")
     plt.show()
```

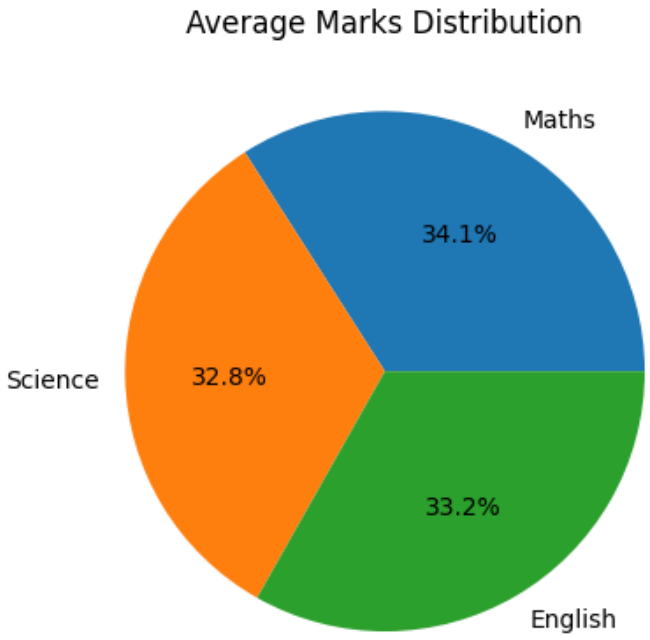


Pie Chart

Average marks distribution.

```
[4]: average = df[["Maths","Science","English"]].mean()

     plt.pie(average, labels=average.index, autopct="%1.1f%%")
     plt.title("Average Marks Distribution")
     plt.show()
```

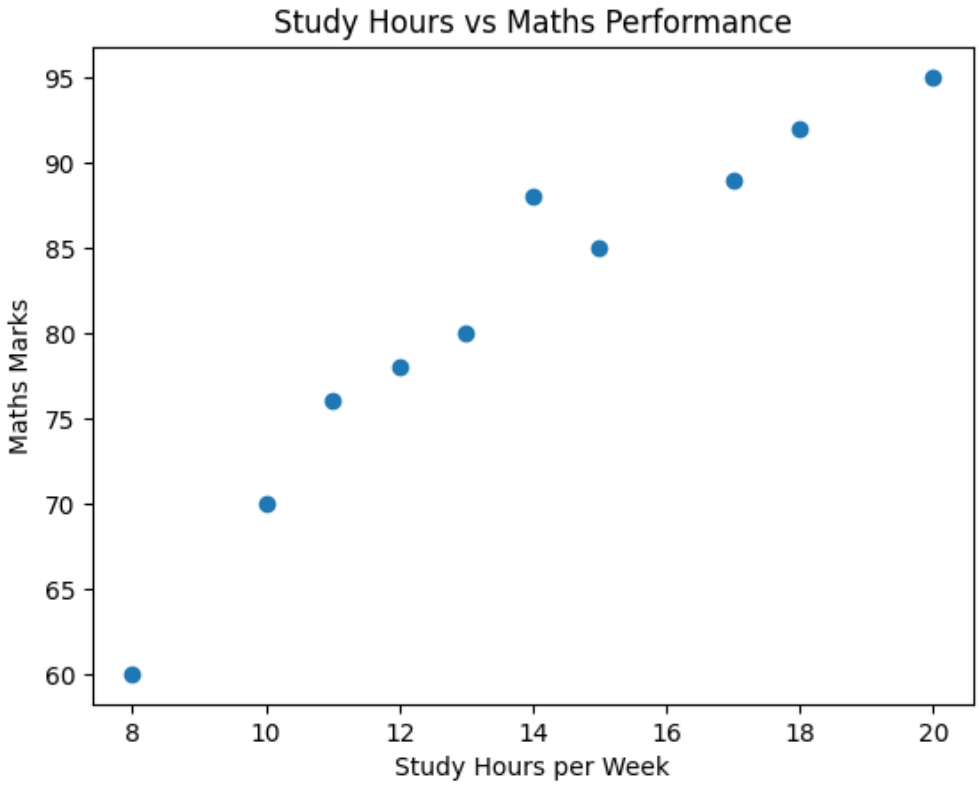


Scatter Plot

Relationship between study hours and maths marks.

```
[5]: plt.scatter(df["Study_Hours"], df["Maths"])

     plt.xlabel("Study Hours per Week")
     plt.ylabel("Maths Marks")
     plt.title("Study Hours vs Maths Performance")
     plt.show()
```

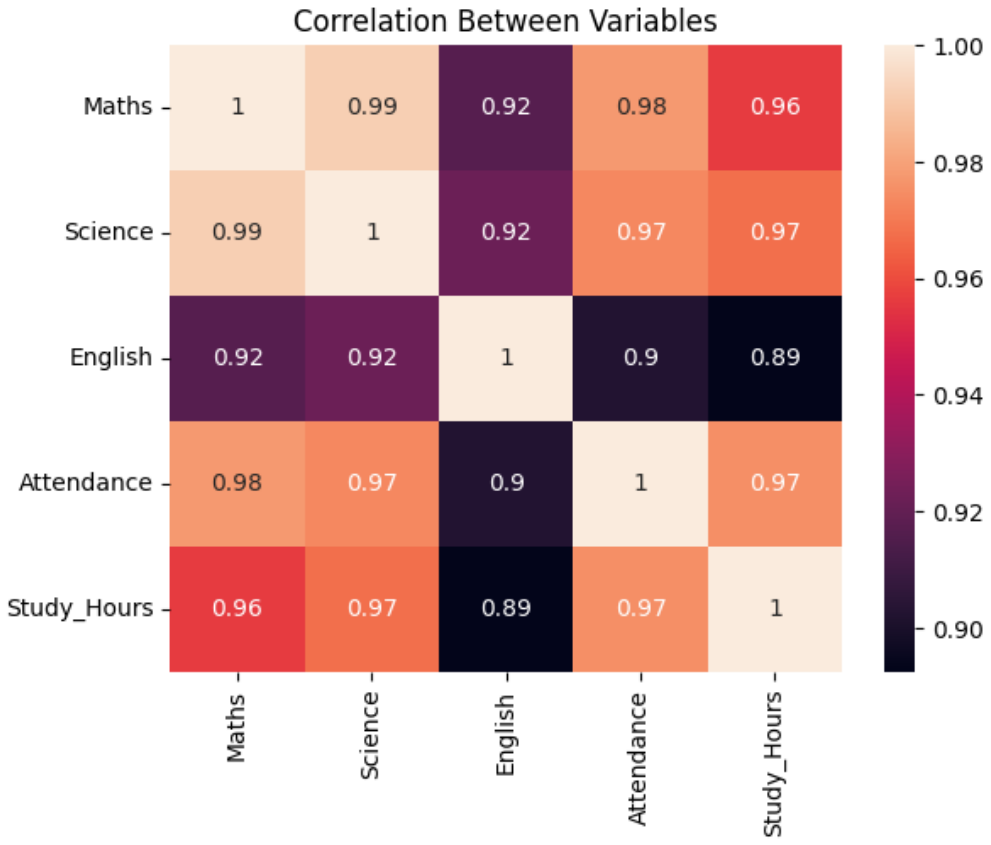


Heat map

Correlation between variables.

```
[6]: sns.heatmap(df.corr(numeric_only=True), annot=True)

     plt.title("Correlation Between Variables")
     plt.show()
```



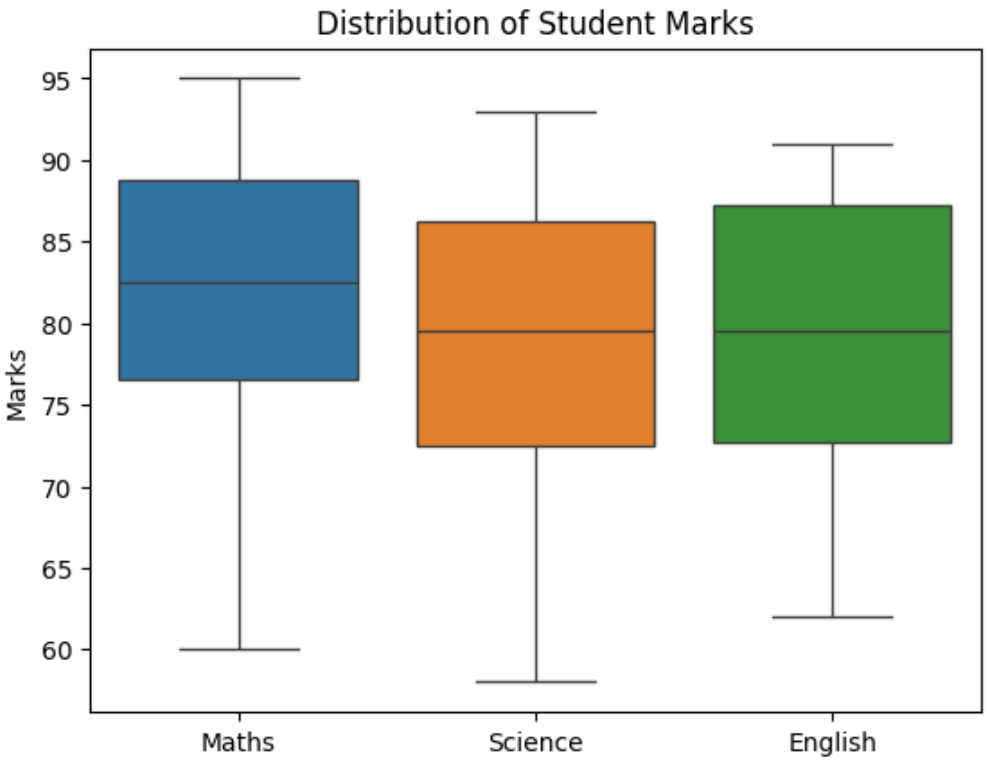
Box Plot

Distribution of student marks.

[7]:
```
import seaborn as sns
import matplotlib.pyplot as plt

sns.boxplot(data=df[["Maths","Science","English"]])

plt.title("Distribution of Student Marks")
plt.ylabel("Marks")
plt.show()
```

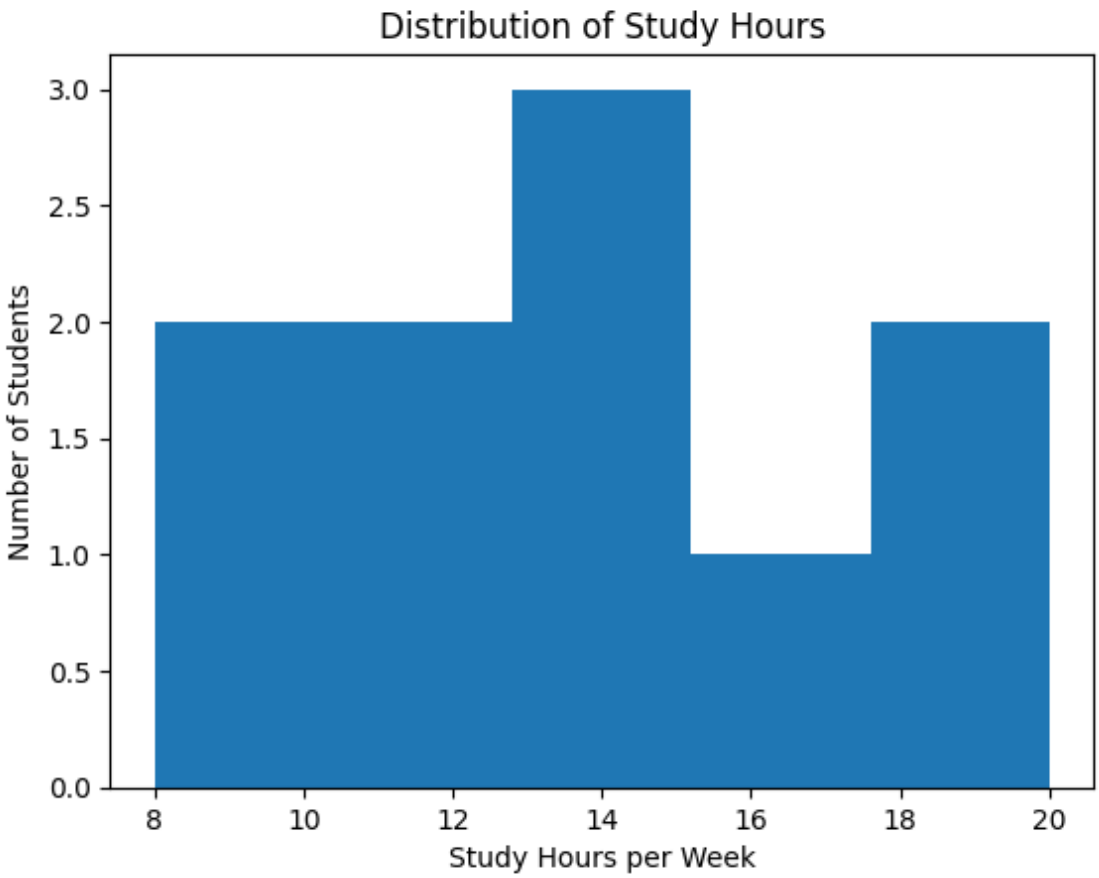


Histogram

Distribution of study hours.

```
[8]: plt.hist(df["Study_Hours"], bins=5)

     plt.title("Distribution of Study Hours")
     plt.xlabel("Study Hours per Week")
     plt.ylabel("Number of Students")
     plt.show()
```



11. Results and Analysis

The analysis of the dataset provided several insights:

- Students with higher study hours generally achieve better marks.
- Attendance has a positive impact on student performance.
- Visualization helps identify trends in academic data.

12. Conclusion

Data visualization helps transform raw data into meaningful insights. Using Python libraries such as Pandas, Matplotlib, and Seaborn makes it easy to analyze and visualize datasets.

This project successfully demonstrates how student performance data can be analyzed and visualized to identify patterns and relationships.

13. Future Scope

This project can be improved by:

- Using larger real-world datasets
- Applying machine learning techniques
- Creating interactive dashboards for data analysis

14. References

- Python Documentation
- Pandas Documentation
- Matplotlib Documentation
- Seaborn Documentation